Name: ________________________ Homework 3 | Math 253 | Huchala

*Due Wednesday of Week 4 at the start of class*

Complete the following problems and submit them in class. 8 points are awarded for thoroughly attempting every problem, and I'll select four problems to grade on correctness. Enough work should be shown that there is no question about the mathematical process used to obtain your answers.

In problems 1–6, determine if the series converges.

1. $\displaystyle\sum_{m=1}^{\infty} \frac{n}{n^2+1}$.

2. $\displaystyle\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$.

3. $\displaystyle\sum_{k=1}^{\infty} \sin(k)$.

4. $\displaystyle\sum_{n=2}^{\infty} \frac{1}{n\sqrt{n^2-1}}$.

5. $\displaystyle\sum_{n=2}^{\infty} \frac{1}{n\ln n}$.

6. $\displaystyle\sum_{m=1}^{\infty} \frac{\ln(m)}{m^2}$.

7. For each of the series in problems 1–6 that converges, find a value of $N$ so that the $N$th remainder $R_N$ is below 0.01. Then compute the $N$th partial sum of the series with a tool like WolframAlpha.